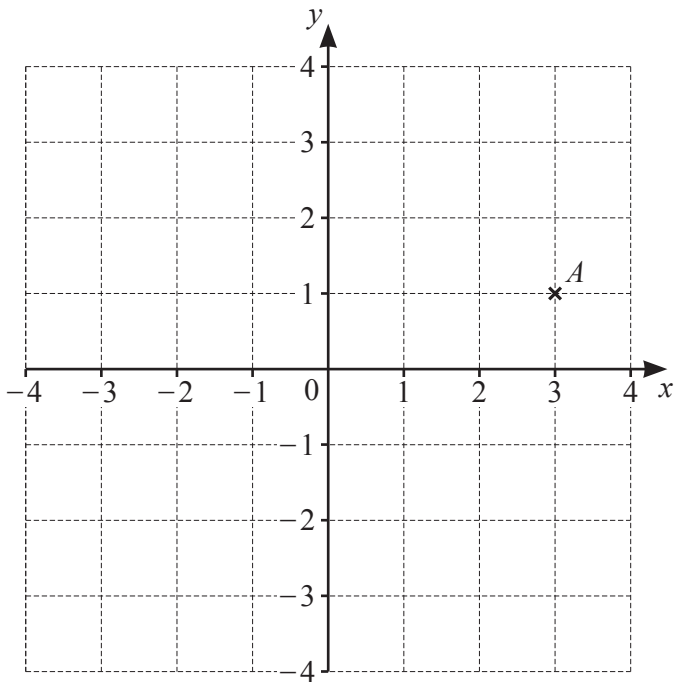
0580_s22_qp_31_Q5

5 **(a)** The grid shows a point A.



(i) Write down the coordinates of point A.

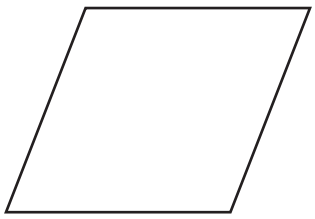
(..... , ...) [1]

(ii) On the grid, plot the point B at $(-1, 3)$. [1]

(iii) C is a point on the grid whose coordinates are whole numbers.

On the grid, mark a point C so that triangle ABC is isosceles. [1]

(b)



The diagram shows a rhombus.

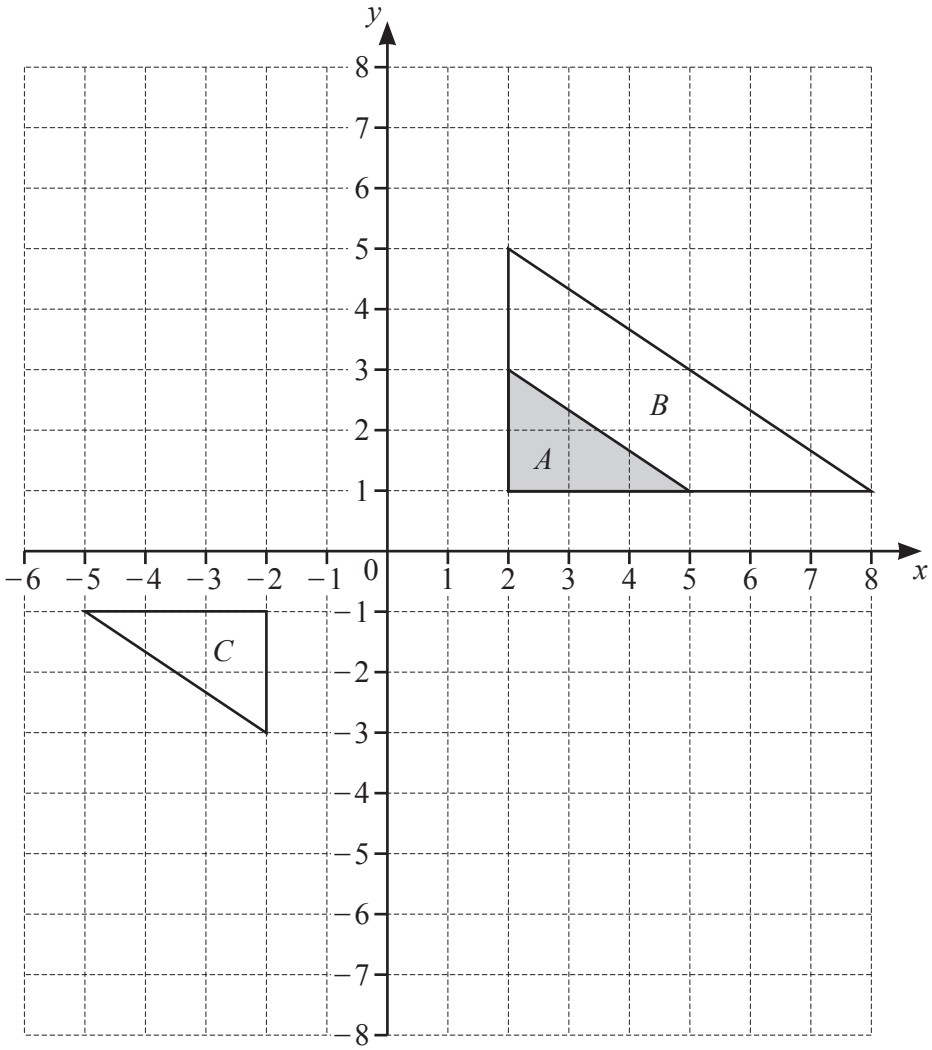
(i) Write down the order of rotational symmetry.

..... [1]

(ii) On the diagram, draw all the lines of symmetry. [2]

0580_s22_qp_31_Q5

(c) The grid shows triangles A, B and C.



(i) Describe fully the **single** transformation that maps triangle A onto triangle B.

.....

..... [3]

(ii) Describe fully the **single** transformation that maps triangle A onto triangle C.

.....

..... [3]

(iii) Draw the image of

(a) triangle A after a translation by the vector $\begin{pmatrix} -5 \\ 3 \end{pmatrix}$, [2]

(b) triangle A after a reflection in the line $y = -2$. [2]